

Batch: A-4

Roll No.: 16010122151

Experiment / Assignment / Tutorial No: 7

TITLE: To understand the importance of card sorting technique for information architecture in UX design process.

Objective: To study and understand the role of card-sorting techniques in UI/UX design.

Upon completion of this laboratory exercise, students will be able to:

1. **Understand** the role of information architecture in UI/UX design process.
2. **Classify** the different card-sorting techniques.
3. **Create** their UI's by demonstrating open & closed card sorting techniques.
4. **Analyze** how the card-sorting techniques give structural presentation of how the data can be visualised.
5. **Appreciate** the value of importance of information architecture with visual hierarchy.

Expected OUTCOME of Experiment: CO 4

Analyze tools and techniques for prototyping and designing applications

Books/ Journals/ Websites referred:

- **Books:**
 - Goodwin, K. (2009). *Designing for the Digital Age: How to Create Human-Centered Products and Services*. Wiley. (Chapters on user scenarios and storyboarding)
 - Lidwell, W., Holden, K., & Butler, J. (2010). *Universal Principles of Design*. Rockport Publishers. (Look for principles related to storytelling and visualization)
 - Saffer, D. (2010). *Designing for Interaction: Creating Smart Applications and Clever Devices* (2nd ed.). New Riders.
- **Journals:**
 - *Interacting with Computers*
 - *International Journal of Human-Computer Interaction*
 - *Personal and Ubiquitous Computing*
- **Websites:**
 - <https://careerfoundry.com/en/blog/ux-design/a-beginners-guide-to-information-architecture/>
 - <https://medium.theuxblog.com/information-architecture-what-is-it-and-where-did-it-come-from-5ba3105fe6b4>
 - <https://www.interaction-design.org/literature/topics/information-architecture>

Background Theory:

Card sorting is a user-centered design (UCD) technique used in information architecture (IA) and UX design to understand how users mentally categorize and organize information. It helps designers create navigation structures, menus, and content groupings that align with user expectations and mental models. The method provides valuable insights into how people perceive relationships between concepts and content items.

In User Experience (UX) design, **Information Architecture (IA)** is the art and science of organizing and labeling content to help users find information and complete tasks. A well-designed IA ensures that users can intuitively navigate a website or application, understand where they are, and predict where they need to go. Poor IA leads to frustration, inefficiency, and users abandoning a product.

Card Sorting is a widely used UX research method for discovering how users group and categorize information. It helps designers understand users' mental models – how they think about and organize content – which is crucial for creating intuitive navigation structures and logical content groupings.

How Card Sorting Works: Participants are given a set of cards, each representing a piece of content or functionality from a website or application (e.g., "Product Catalog," "Customer Support," "My Orders," "About Us," "Contact Information"). They are then asked to group these cards in a way that makes sense to them.

Types of Card Sorting:

1. Open Card Sorting:

- **Method:** Participants sort cards into groups that make sense to them and then create their *own* labels for each group.
- **Purpose:** Best for exploring how users naturally categorize content and for generating new category names. It helps discover unforeseen relationships between content items.
- **When to use:** Early in the design process, when you need to understand users' mental models and generate ideas for primary navigation or site structure.

2. Closed Card Sorting:

- **Method:** Participants sort cards into pre-defined categories that you provide.
- **Purpose:** Best for evaluating an existing information architecture or testing a proposed structure. It helps validate if users understand and agree with your pre-defined categories.
- **When to use:** When you have an existing structure or a set of proposed categories and want to see if users can successfully sort content into them.

Benefits of Card Sorting:

- **User-Centric IA:** Ensures that the information architecture reflects how users think, not just how the organization thinks.
- **Improved Findability:** Helps users locate information more easily, reducing frustration.
- **Reduced Development Costs:** Identifying optimal structures early prevents costly redesigns later.
- **Better User Experience:** Leads to more intuitive navigation and a more satisfying overall experience.
- **Validation:** Provides data-driven insights to support design decisions.

Analysis of Results: After conducting a card sort, the data is analyzed to find common patterns. Tools can generate dendrograms (tree diagrams showing clustering) or similarity matrices to visualize how frequently items were grouped together. This analysis helps identify strong groupings and potential areas of confusion.

This lab will provide hands-on experience with conducting and analyzing a card sorting exercise, demonstrating its practical application in designing user-friendly information architectures.

Why Storyboard in UI/UX Design?

Card sorting is used in UI/UX design to help designers understand how users think about and categorize information. It plays a crucial role in building **intuitive navigation structures** and **user-friendly interfaces**. The primary purpose is to align the system's information architecture with the **mental models** of its users.

1. Understand Users' Mental Models

Every user has a unique way of grouping and labeling information. Card sorting helps reveal these **mental models**—the internal maps that users create to make sense of digital content.

- By analyzing how users group items, designers can understand how users expect information to be organized.
- This ensures that menus, categories, and labels match the user's thought process, reducing confusion and cognitive effort.

2. Improve Information Architecture (IA)

A major reason for conducting card sorting is to **design or refine the information architecture** of a website or application.

- It identifies logical groupings of content or features.
- The results guide how sections, menus, and submenus should be structured.
- This improves the **findability** and **usability** of content.

3. Enhance Navigation and User Flow

Poor navigation is one of the most common causes of user frustration.

- Card sorting helps determine how users expect navigation to flow.
- It ensures that important information is easy to locate, improving the overall **user experience (UX)** and reducing time spent searching.

4. Validate Design Decisions

Before building or redesigning an interface, designers can use **closed card sorting** to validate whether existing categories make sense to users.

- This provides **data-driven evidence** for organizing content.
- It prevents costly design changes later in the development process.

5. Support User-Centered Design (UCD)

Card sorting is a **participatory** and **user-centered** method. It involves end users directly in the design process, ensuring that decisions are made based on **user input rather than designer assumptions**.

- This results in interfaces that feel natural, familiar, and efficient for the target audience.

6. Identify Naming and Labeling Conventions

Users often use different words or terms to describe the same concept.

- Through card sorting, designers discover the **language users naturally use**.
- This helps in creating **clear, consistent, and meaningful labels** that improve comprehension and engagement.

7. Cost-Effective Early Testing Method

Card sorting is inexpensive and easy to conduct—even with small groups of participants or using online tools.

- It can be performed early in the design phase, providing quick feedback.
- The insights gained prevent larger usability issues later.

Laboratory Activities

Duration: 3-4 hours (including scenario development, storyboarding, and documentation)

Materials:

- Computer with internet access
- Access to a digital whiteboarding tool (e.g., Miro, Mural - free tiers are often sufficient) or a dedicated storyboard creation tool (e.g., StoryboardThat). Alternatively, paper and drawing materials can be used for hand-drawn storyboards.
- Notebook or digital document for planning and reflection.

Activity:

Choose the scenarios based on completed wireframes. You will demonstrate the open and closed card sorting techniques for the application under consideration.

Procedure:

1. Scenario & Content Selection (45 min):

- **Choose a product/website under consideration:** Select the product UI under design for which you will design/improve the information architecture.
 - *Examples:* A university department website, a small online bookstore, a local community events portal, a recipe sharing app.
- **Identify 20-30 distinct content items/features:** Based on your chosen product, brainstorm a list of 20-30 individual pieces of content or functionalities that would typically be found on such a site/app. Each item should be concise and represent a single concept.
 - *Examples for a university department website:* "Course Catalog," "Faculty Directory," "Admissions Requirements," "Research Labs," "Student Resources," "Alumni News," "Contact Us," "Events Calendar," "Apply Now," "Scholarships."
- **Prepare your "cards":** Create a list of these items in a spreadsheet. Each item will become a "card" for the sorting exercise.

2. Card Sorting Method Selection & Setup (45 min):

- Decide whether you will conduct an **Open Card Sort** or a **Closed Card Sort**. For this introductory lab, an **Open Card Sort** is highly recommended as it provides richer insights into user mental models.
- **If using an online tool (Recommended):**
 - Sign up for a free trial or student account on a platform like OptimalSort.
 - Set up your card sort project, inputting your 20-30 content items as cards.
 - Prepare clear instructions for participants (e.g., "Group these cards in a way that makes sense to you. Label each group with a name you think is appropriate.").
- **If conducting a manual card sort:**
 - Write each content item clearly on a separate index card.
 - Prepare a quiet space for the participant(s).
 - Have a method to record the groupings (e.g., take photos, write down the groups).

3. Conducting the Card Sort (60-90 min):

- **Recruit 2-3 participants:** These can be classmates, friends, or family members. Explain the purpose of the exercise (to understand how they organize information, not to test them).
- **Provide instructions:** Clearly explain the task based on your chosen method (open or closed).
- **Observe and record:**
 - **For online tools:** The tool will automatically collect the data.
 - **For manual sorts:** Observe their process. Note any hesitation, questions, or comments they make. After they finish, record their groupings and category names.
- **Debrief (Optional but Recommended):** After each participant finishes, ask them briefly about their thought process, why they grouped certain items together, and if any items were particularly difficult to categorize.

4. Data Analysis & Interpretation (60-90 min):

- **For online tools:**
 - Utilize the analysis features provided by the tool (e.g., similarity matrix, dendrogram, participant agreement matrix).
 - Identify the most common groupings and category names. Look for strong agreements and outliers.
- **For manual sorts:**
 - Manually compile the results from all participants. Create a table showing which cards were grouped together by each participant.
 - Look for patterns: Which cards were frequently grouped together? Which category names appeared most often for similar groupings?
 - Calculate simple agreement metrics (e.g., how many participants grouped "Contact Us" and "About Us" together).
- **Formulate IA Recommendations:** Based on your analysis, propose a revised information architecture for your hypothetical product/website. This should include:
 - A proposed set of main navigation categories.
 - Which content items would fall under each category.
 - Justification for your proposed structure based on the card sorting results.

Card-Sorting:

1) Movies

Card Sort

IMDB entertainment site

Hi , welcome to your Cosort Dashboard

Step 1:
Click on a card

Step 2:
Drag and drop it into the category which seems appropriate

Step 3:
After sorting all the cards, click on "Done" button to complete the session

Movies	TV	Community	News	Videos	Apps
Top 500	TV Land	Message Board	Top News	Search Videos	Windows Phone app
In Theaters	TV Home	My ratings	Movie news	Browser addons	iPhone app
Featured Movie		Watchlist	Celebrity news		Android Tablet app
Top movies	Coming soon	Favorite lists			
All trailers		Contributor zone			
Trailers		Private			
		Games			

Sorting finished! Hit Done to proceed.

100%

SESSION PROGRESS

Done

Card Sort Hi, welcome to your Cosort Dashboard

Below is your card sorting results against the information architecture of Dell website. You can see both the information architectures.
Note: This is not to show where you were wrong but its your individual mental model as a user of this website.

Your card sort result

Movies	TV	Community
Top 250	TV Listing	Message Board
In Theaters	TV Home	My ratings
Featured News	Coming soon	Watchlist
HD Trailers		Newest lists
Top movies		Contributor zone
Tickets		Trivia app
		Genre

News	Videos	Apps
Top news	Browse Videos	Windows Phone app
Movie news	Search Videos	iPhone app
Celebrity news		Android Tablet app

Information architecture of the website

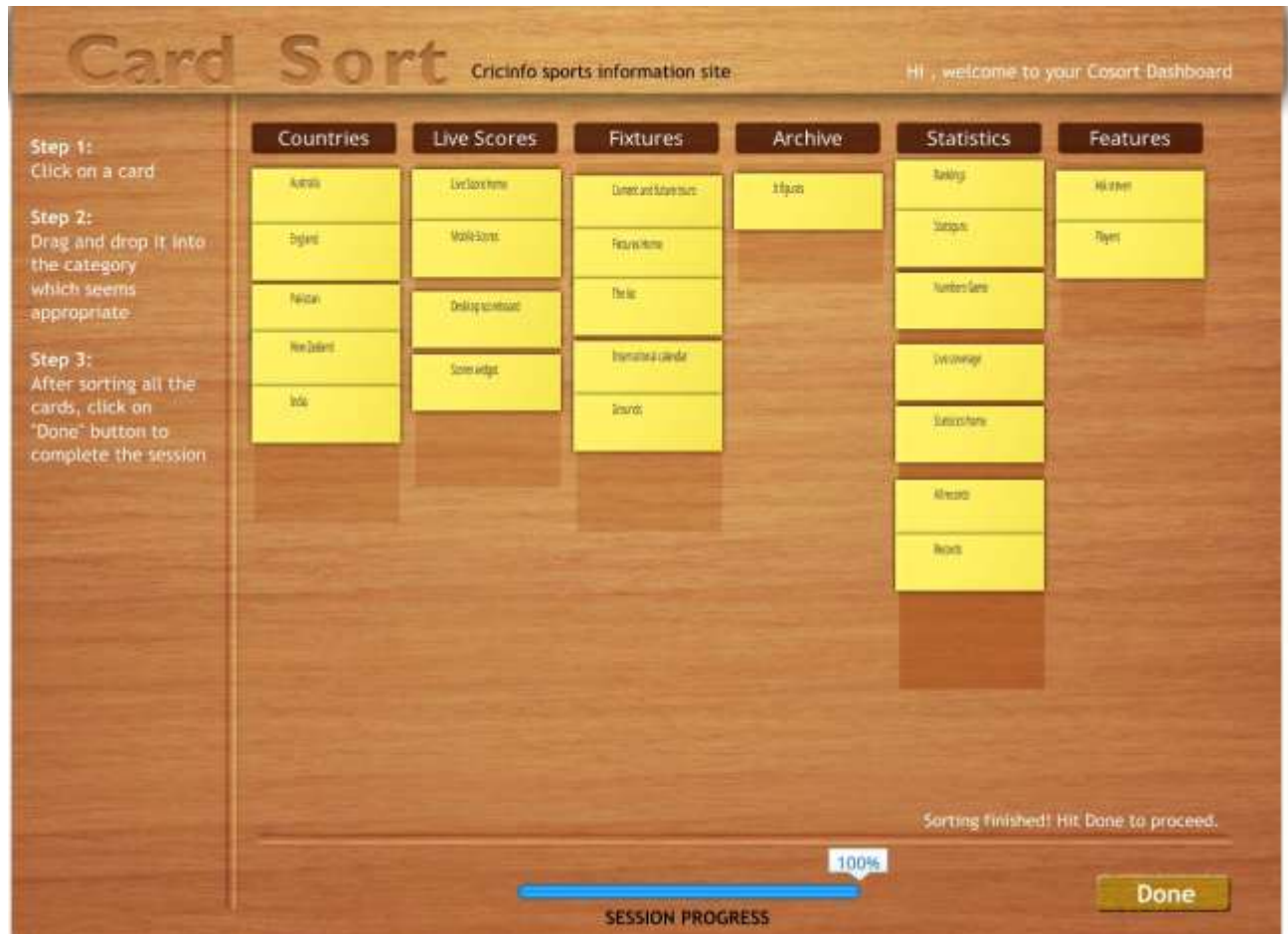


Movies	TV	Community
In Theaters	TV Home	Message board
Coming soon	TV Listings	Newest lists
Genre		My ratings
Top Movies		Contributor zone
Watchlist		
Tickets		
Top 250		

News	Videos	Apps
Top news	Search Videos	Android tablet app
Movie news	HD Trailers	app
Celebrity news	Browse videos	Trivia App
Featured news		iPhone App
		Windows phone app

Choose another card deck Exit

2) ESPNCricInfo



Card Sort Hi, welcome to your Cosort Dashboard

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Note: This is not to show where you were wrong but its your individual mental model as a user of this website.

Your card sort result

Countries	Live scores	Fixtures
Australia	Live Score home	Current and future tours
England	Mobile Scores	Fixtures Home
Pakistan	Desktop scoreboard	The list
New Zealand	Scores widget	International calendar
India		Grounds

Archive	Statistics	Features
It figures	Rankings	Ask steven
	Statsguru	Players
	Numbers Game	
	Live coverage	
	Statistics home	
	All records	
	Records	

Information architecture of the website

Countries	Live scores	Fixtures
Australia	Live scores	Fixtures home
India	home	Live coverage
England	Desktop-scoreboard	Current and future tours
New Zealand	Scores widget	International calendar
Pakistan	Mobile scores	

Archive	Statistics	Features
Players	All records	Regulars
Grounds	Numbers-game	Highlights
Rankings	The list	Opinions
	Statistics Home	
	Statsguru	

Choose another card deck Exit

3) Dell Website

Card Sort

Dell Computer website

Hi, welcome to your Cosort Dashboard

Step 1:
Click on a card

Step 2:
Drag and drop it into the category which seems appropriate

Step 3:
After sorting all the cards, click on "Done" button to complete the session

Company Info	Shop For	Products	My account	Participate	Support
About	Large enterprise	Workstations	Sign in/register	Share your ideas	Customer complaints
Newstroom	Small/medium enterprise	Laptops	Order status	Real-time blog	Product Support
Investors	Policymaker	Desktops		Join the Dell ecosystem	Drivers and downloads
Careers	Home and home office	Peripherals		View ratings	All Support Options
Corporate responsibility		Mobile devices			

Sorting finished! Hit Done to proceed.

100%

SESSION PROGRESS

Done

Card Sort

Hi, welcome to your Cosort Dashboard

Below is your card sorting results against the information architecture of Dell website. You can see both the information architectures.
Note: This is not to show where you were wrong but its your individual mental model as a user of this website.

Your card sort result

Company Info	Shop For	Products
About	Large enterprise	Workstations
Newsroom	Small/medium enterprise	Laptops
Investors	Public sector	Desktops
Careers	Home and home office	Printers
Corporate responsibility		Mobile device

My account	Participate	Support
Sign in/register	Share your ideas	Customer complaints
Order status	Read our blog	Product Support
	Join the discussion	Drivers and downloads
	View videos	All Support Options

Information architecture of the website

Company info ▼	Shop For ▼	Products ▼
About	Small/medium	Laptops
Investors	Enterprise	Printers
Careers	Home and home office	Workstations
Corporate responsibility	Public Sector	Desktops
Newsroom	Large Enterprise	Mobile device

My account ▼	Participate ▼	Support ▼
Sign in/Register	Join the discussion	All support options
	Share your ideas	Drivers and Downloads
	View videos	Product support
	Read our blog	Customer Complaints

Choose another card deck Exit

Card-sorting for E-commerce website Flipkart:

Flipkart Navigation Menu	
SECTION	CATEGORIES & DETAILS
Shop By Category	Electronics → Mobiles, Laptops, TVs, Cameras Fashion → Men, Women, Kids Home & Furniture → Furniture, Decor, Kitchen Appliances → AC, Washing Machines, Refrigerators Grocery → Food, Beverages, Household Items
Deals & Rewards	SuperCoins, Top Offers, Coupons, Gift Cards, Plus Zone
My Account	Orders, Wishlist, My Chats, My Addresses, Notifications, Account Settings
Support	Help Center, Returns & Refunds, Order Tracking, Payments, Report an Issue
Explore & Seller Zone	Become a Seller, Advertise, Careers, Flipkart Stories, Press Releases

Post Laboratory Questions:

Following questions are to be handwritten and attach photograph of the answer in this document.

1. **Method Justification:** You chose either an Open or Closed Card Sort for this lab. Justify your choice based on the objectives of the lab and the type of insights you aimed to gather. If you had chosen the other method, how would your approach and the expected outcomes have differed?

An Open Card Sort was selected for this lab because it allowed participants to freely organize and label items without being confined to predefined categories, making it ideal for uncovering users' mental models and identifying natural grouping patterns within app content. This approach was aimed at gaining deeper qualitative insights and identifying new or unexpected item groupings. Had a Closed Card Sort been chosen instead, users would have been limited to preset categories, which might validate an existing structure but would restrict the opportunity to learn how users genuinely perceive and categorize information. Therefore, Open Card Sort was preferable for early exploration, whereas Closed Card Sort would be better suited for validating a finalized structure.

2. **Key Findings from Analysis:** Describe the most significant patterns or insights you discovered from analyzing your card sorting results (e.g., a strong consensus on a particular grouping, an unexpected grouping, a confusing card). How did these findings influence your proposed information architecture?

The analysis surfaced several notable trends. Many users consistently grouped items like phones, laptops, and headphones together as "Electronics," showing a strong consensus for high-level categories. However, some participants surprised by blending items such as shoes and watches under "Accessories" instead of "Fashion," reflecting different views on category relationships. Ambiguities arose around items like gift cards or customer support, with users unsure of their appropriate placement, revealing the need for clearer labeling. These findings directly shaped the recommended information architecture, reinforcing some top-level categories and suggesting redesigns for confusing sections to improve clarity and usability.

3. **Impact on User Experience:** How does the process of involving users in categorizing content (via card sorting) directly contribute to a more intuitive and user-friendly experience in the final product? Provide a specific example from your lab where user input helped clarify or improve the organization of content.

User involvement in content categorization helped ensure that the final structure closely matched how users think, making navigation easier and reducing mental effort. For instance, input from the sorting exercise showed that users consistently grouped items like offers, discounts, and coupons under a unified "Deals" category. Initially, these were spread across several menus, but reorganizing based on user feedback made promotions easier to find and understand, marking a clear improvement in usability and user satisfaction.

4. **Limitations and Next Steps:** Discuss one significant limitation or challenge you encountered while conducting or analyzing your card sorting exercise. What steps would you take next in a real-world UX design project to further validate or refine the information architecture based on these card sorting results?

A key challenge was the small number of participants, which means the findings may not capture the full diversity of the intended audience. Additionally, some users interpreted card labels differently, leading to inconsistent groupings. In a real-world UX project, next steps would include running a Closed Card Sort or tree testing with a larger and more varied sample to validate the structure, refining unclear labels based on observed confusion, and iterating on the architecture before retesting with users to ensure the design is robust and intuitive.